Lab 4 — Instructor Key: Bending

For instructors only. Do not distribute to students. Every number in this key is computed by `src/lab04_analysis.py` (and `src/make_key.py`) from `sample_data.csv` and `src/specimen_measurements.json`, or by the simulation's own model code run in Node. To build a key for your own class data, put your CSV and measurements in the same format and run `python3 src/lab04_analysis.py your.csv your_measurements.json`, then `python3 src/make_key.py`. Uncertainties are expanded ($U = 2u$) unless marked u. Sign convention as in the handout (Beer 8e): V and M as in Fig. 5.6, $M > 0$ sagging, $\sigma_x = -My/I$ with y up from the neutral axis, tension positive.

1. Results at a glance (sample data)

Table K1. Results from the sample data.

Quantity	Measured	Reference or prediction	Verdict (Eq. 18)
Bar A: b, h (mm); I (10^5 mm^4)	25.420 ± 0.026, 50.780 ± 0.026; 2.7738 (u_I = 0.092 %)	25.4 × 50.8 nominal	—
Shear span a (mm)	200.00 ± 0.41	200	—
V/P in the left shear span (A1)	0.5005 ± 0.0242	0.5 (Eq. 2)	agree
Neutral axis, bar A (mm above bottom)	25.392 ± 0.026	$h/2$ = 25.390	agree
E of 6061-T6 from M–κ (GPa)	68.39 ± 0.90	Lab 2: 68.46 ± 0.94	agree
ν from anticlastic strain	0.3294 ± 0.0059	Lab 2: 0.330 ± 0.006	agree
Neutral axis, bar B (mm above bottom)	23.985 ± 0.039	Eq. 9: 23.95 ± 0.18	agree
EI, bar B (N·m²)	38 606 ± 64	$E_1 I_t$: 38 751 ± 808	agree
M at 40 kN and κ = 0.2935 m^{-1} (N·m)	4 000	EPP 4 054; material curve 4 003	EPP +1.34 %; curve +0.07 %
Flexure stress vs actual at 40 kN (MPa)	actual 282.0	Mc/I = 366.2	overstated 29.8 %
Permanent curvature (m^{-1})	0.0827	Eq. 14: 0.0826 ± 0.0028	agree
Straightedge gap after Part C (mm)	2.85 ± 0.05	middle 200 mm only: 2.48 ± 0.09; whole bar from the measured M–κ: 2.87	gap 15 % larger: shear spans yielded too
D1 lightest acceptable section	steel I-120, aluminum I-160	—	—

The sample data were generated from a known "truth" model (`src/generate_sample_data.py`, values in `src/truth.json`): plane sections, 6061-T6 following Lab 1's Ramberg–Osgood curve (E = 68.9 GPa), the steel strip elastic, a common +0.5 % gage-factor error, true gage heights that differ from the recorded ones by a few hundredths of a millimeter, load points 0.4 mm inside their nominal positions, and 1 με reading noise. Because every strain reads 0.5 % high, the measured E is 0.5 % low: correct for the gages, invisible to the student, and inside the stated uncertainty.

The closure test in `tests/test_lab04_independent.py` confirms that the analysis recovers the truth values within their stated uncertainty.

2. Pre-lab solutions

P1. (a) $I = bh^3/12$ = 2.7749 × 10^5 mm^4, S = 10 925 mm³. At 20 kN, $M = (P/2)a$ = 2 000 N·m, $\sigma_m = M/S$ = 183.1 MPa, $\varepsilon = \sigma/E$ = 2 674 με, $\kappa = M/(EI)$ = 0.1053 m^{-1}, ρ = 9.50 m. (b) $M_Y = \sigma_Y S$ = 2 990 N·m, so $P_Y = 2M_Y/a$ = 29.90 kN; $M_p = 1.5M_Y$ gives 44.85 kN. Part C's 40 kN is 1.338 P_Y and 0.892 P_p: well into the plastic range but short of collapse. (c) At 2.5 kN the bottom strain is about 334 με, so 1 με is 0.30 %. (d) $V/(bh)$ = 7.75 MPa, 4.2 % of the bending stress (the parabolic maximum, 11.6 MPa, is Lab 5's subject). Units: N·m/mm³ → MPa after ×10³; με = 10^{-6}.

P2. (a) Figure 2: $V = P/2$, 0, $-P/2$; integrating, M rises to $Pa/2$, stays flat, and falls by $Pa/2$ to zero at L. (b) $a_1 + a_2 = L$, so the loads are still symmetric: $R_A = R_B = P/2$, and $M = (P/2)a_1$ = 100.200 × 10^{-3} P (m), 0.200 % above Eq. 2. The middle is still in pure bending. (c) R_A = 0.50033 P and R_B = 0.49967 P (Appendix D.1); between the loads $V = R_A - P/2$ = 0.000333 P, so M changes by 0.067 % across the middle third. Strictly not pure bending, but far below anything the gages can see. *Lesson:* the measurement is insensitive to small load-point errors because the moment in the middle depends on a only to first order.

P3. $n = 205/68.46$ = 2.9944. With y from the bottom of the strip, transformed areas nbt_s at $t_s/2$ and bh at $t_s + h/2$: $\bar{y}$ = 23.967 mm and I_t = 5.6607 × 10^5 mm^4 (bar alone: 2.7749 × 10^5 mm^4; ratio 2.040). At 15 kN, M = 1 500 N·m: steel bottom +190.2 MPa; interface aluminum +46.68 MPa and steel +139.8 MPa (ratio n); aluminum top −87.9 MPa. (b) $t_s \to 0$: $\bar{y} \to h/2$ and $I_t \to bh^3/12$. $n = 1$: a homogeneous bar of depth $h + t_s$, with $\bar{y} = (h + t_s)/2$.

P4. (a) M_Y = 2 990 N·m, $\kappa_Y = \sigma_Y/(Ec)$ = 0.1574 m^{-1}, M_p = 4 485 N·m. (b) M/M_Y = 1.3378; Eq. 12 gives $(\kappa_Y/\kappa)^2 = 3(1 - M/1.5M_Y)$, so $\kappa_Y/\kappa = y_Y/c$ = 0.5696, κ = 0.2763 m^{-1}, and the elastic core is ±14.47 mm. (c) Elastic recovery $M/(EI)$ = 0.2106 m^{-1}, so κ_p = 0.0657 m^{-1}. Residual stress: top $-\sigma_Y + Mc/I$ = −273.7 + 366.1 = +92.4 MPa (tension), bottom −92.4 MPa. (d) $-\int y\sigma_{res}\, dA = M - M = 0$ (Appendix D.7); students may also integrate the piecewise-linear residual distribution numerically.

P5. Simulation, presets as shipped (nominal section, E = 68.9 GPa). Part A: M = 2 000 N·m, largest $|\sigma|$ = 183.1 MPa. Part B: elastic NA 24.004 mm, I_t = 5.6472 × 10^5 mm^4 (slightly different from P3 because the page uses E = 68.9 GPa). Part C at κ = 0.292 m^{-1}: material curve 4 008 N·m, EPP 4 060 N·m (+1.28 %). The EPP law is larger: aluminum's curve bends over before σ_Y (the 0.2 % offset point), so the outer fibers carry less than σ_Y just past yield, and Part C's strains are not yet large enough for the hardening to make it up. "Overload, then unload" ($\kappa = 3\kappa_Y$ = 0.4697 m^{-1}, M = 1.4444 M_Y): permanent curvature 0.2435 m^{-1}.

P6. (a) Pinch points at the rollers and spreader (hands clear when the crosshead can move); stored elastic energy if the bar shifts or a roller slips (bar centered and square, stop on a load drop); overloading the bar or fixture (overload stop set per part); manual handling of 2.5 kg bars. (b) Part C bends the bar permanently, so it must come after Part A, which needs an undamaged bar; an overload stop set before the plastic part is the only protection against running past 40 kN toward collapse at about 44.85 kN, where the deflection would grow quickly.

3. Analysis solutions

A1. Slopes of bottom strain against P converted with $M = ES\varepsilon$ (E from A3): M/P = 49.91 mm at x = 100 mm (statics 50.0, −0.18 %), 74.94 mm at 150 mm (75.0, −0.08 %) and 100.06 mm at midspan (100.0, 0.06 %); each ±0.67 % (u, mostly from E). V/P **= 0.5005 ± 0.0242**, $|\Delta|$ = 0.0005 ≤ 0.0242: agree. *Check:* the line through the two shear-span points meets $M = 0$ at x = 0.28 mm, at the support, as a pin must require. *Interpret:* the slope of the moment diagram is the reaction ($V = R_A$ in the shear span); between the loads both load points' halves cancel, so the slope, V, is zero. The shear is a difference of two nearly equal moments, which is why its relative uncertainty (about 2.4 %, u) is several times that of M.

A2. 11 steps above 1 kN. Curvature: 0.02637 m^{-1} at 5 kN, 0.05268 at 10 kN, 0.10540 at 20 kN (linear in load). RMS residuals 0.79–3.10 με (1.81 at 20 kN). At 20 kN, top −2 675 με and bottom 2 677 με. y_{NA} **= 25.392 ± 0.026 mm** vs $h/2$ = 25.390 mm: difference 0.002 mm, within 0.026. *Check:* residuals at the 1–3 με level with no curvature in the profile; equal and opposite extreme strains. *Interpret:* a curved profile would mean sections warp (or a gage is misplaced); residuals growing with load in the same pattern would mean gage-height errors (C3), not physics.

A3. 11 loading steps; slope EI = 18 971 N·m². E **= 68.39 ± 0.90 GPa** vs Lab 2 68.46 ± 0.94: $|\Delta|$ = 0.07 ≤ 1.30, agree. Budget (u, %): fit 0.031, b 0.051, $3u_h/h$ 0.076, a 0.102, load cell 0.289, gage factor 0.577; total 0.661 %. At 20 kN: flexure formula 183.0 MPa vs $E\varepsilon$ = 183.1 MPa (ε = 2 677 με). *Check:* slope units N·m/(m^{-1}) = N·m²; $I \propto h^3$ so 1 % in h is 3 % in E. *Interpret:* the gage factor dominates. A common gage-factor error scales every strain, so it scales κ and biases E, but it cancels in $y_{NA} = -A_0/A_1$ (C3).

A4. ν **= 0.3294 ± 0.0059** vs 0.330 ± 0.006: agree. At 20 kN ρ = 9.49 m, so $\rho' = \rho/\nu$ = 28.8 m. *Check:* the compressed top widens, so the transverse gage there reads positive (882 με at 20 kN). *Interpret:* the anticlastic curvature makes the top face slightly saddle-shaped; near the edges the transverse strain is affected by the free edges and the gage's finite length, so a centered gage is best (§4.3 assumes the section is free to deform).

A5. Measured **NA 23.985 ± 0.039 mm** vs Eq. 9 with measured dimensions and n = 2.9974: **23.95 ± 0.18 mm**; $|\Delta|$ = 0.033 ≤ 0.186, agree (the prediction's uncertainty is dominated by E_{steel}). All-aluminum bar of the same depth: 28.57 mm. EI **= 38 606 ± 64 N·m²** vs $E_1 I_t$ = **38 751 ± 808** (−0.38 %), agree. At 14.993 kN (M = 1 499.3 N·m), from the fitted strain line (RMS 1.36 με): steel bottom 190.8 MPa (predicted 190.0, 0.41 %), aluminum interface 46.8 (46.6), steel interface 140.1 (139.5), aluminum top −88.1 (−87.8). $\int \sigma\, dA$ = 0.17 % of the compressive resultant. *Check:* it must vanish in pure bending (first of Eqs. 4); the small remainder is the strain-line fit and the 0.5 % gage-factor bias, which cancels in ratios. *Interpret:* the composite is 2.035 times as stiff as bar A (18 971 N·m²); the steel's bottom fiber is the most highly stressed point, about twice the aluminum's top.

A6. Table K2 lists the Part C steps. M_Y = 2 990 N·m, M_p = 4 485 N·m, κ_Y = 0.1576 m^{-1} (measured E, Lab 1 σ_Y).

Table K2. Part C (loading branch and unloading); model moments at the measured curvature.

P (kN)	M (N·m)	κ (m^{-1})	EPP, Eq. 12 (N·m)	Material curve, strips (N·m)	y_{NA} (mm)	RMS (με)
5.0	501	0.0264	500	500	25.40	0.3
10.0	1 000	0.0526	999	999	25.40	2.0

P (kN)	M (N·m)	κ (m^{-1})	EPP, Eq. 12 (N·m)	Material curve, strips (N·m)	y_{NA} (mm)	RMS (με)
15.0	1 501	0.0791	1 500	1 500	25.40	2.2
20.0	2 000	0.1054	1 999	1 999	25.40	2.3
25.0	2 501	0.1323	2 509	2 499	25.39	3.9
28.0	2 801	0.1500	2 846	2 800	25.39	4.1
31.0	3 100	0.1712	3 218	3 100	25.39	4.5
34.0	3 400	0.1986	3 544	3 401	25.39	4.8
37.0	3 700	0.2365	3 821	3 702	25.39	5.7
40.0	4 000	0.2935	4 054	4 003	25.39	6.6

At 40 kN (M/M_Y = 1.338): measured κ = 0.2935 m^{-1}, extreme strain 0.745 %. EPP predicts 4 054 N·m (+1.34 %), the strip summation with Lab 1's curve 4 003 N·m (+0.07 %). **Flexure stress 366.2 MPa vs actual 282.0 MPa: 29.8 % too high. Permanent curvature: measured 0.0827 m^{-1}, Eq. 14 0.0826 ± 0.0028: agree** (elastic recovery 0.2109 m^{-1}). The zero-load surface strains, ±2 099 με, are $\kappa_p c$. **Straightedge gap: measured 2.85 ± 0.05 mm; Appendix D.9 with κ_p confined to the middle 200 mm gives 2.48 ± 0.09 mm.** The difference, 0.37 mm (limit 0.10), is real: the moment exceeds M_Y in the shear spans beyond $x = M_Y a/M_{max}$ = 149.5 mm, so the bar also yielded near the load points. Integrating $\kappa_p(x) = \kappa(M(x)) - M(x)/(EI)$ along the whole bar, with $\kappa(M)$ interpolated from the measured loading curve, gives 2.87 mm, in agreement. (Expect only the first estimate from most students; the second is worth credit as an interpretation.) Surface residual stress on the tension side (strip summation, material curve): **−84.2 MPa**. *Check:* y_{NA} = 25.39 mm at 40 kN and the RMS residual 6.58 με vs 1.81 at 20 kN in Part A. Per unit curvature, 22.4 vs 17.2 με·m: the residual grows with κ in the same pattern, the signature of gage-height errors (C3), so plane sections still hold. *Interpret:* aluminum's curve bends over before the 0.2 % offset point, so near yield the outer fibers carry less than σ_Y and EPP overpredicts M. The same gradual yielding means the measured curvature at 4000 N·m is larger than EPP's 0.2768 m^{-1}, so the permanent set (0.0827) is 1.25 times the bilinear EPP estimate (0.0659 m^{-1}). Also worth credit: the transverse-to-longitudinal strain ratio on the top face rose to 0.408 at 40 kN (plastic flow is incompressible, so the ratio heads toward 0.5); the bottom gage at x = 150 mm, where $M \approx M_Y$ at 40 kN, kept 138 με of 4 144 με: the plastic zone reached just to it, consistent with the gap.

A7. Use Table K1. Expected verdicts: every Table 5 row agrees except two model-form discrepancies. The moment at 40 kN: EPP is +1.34 % (the comparison's U is about 1 %), and the material curve removes it. The straightedge gap: the middle-only estimate of D.9 is 15 % low because the shear spans yielded too (A6). Full credit needs each discrepancy above U explained with a named, quantified cause.

M1. (a) Flexure formula vs actual extreme-fiber stress (simulation, material curve): 30 kN: 275 vs 252 MPa (9 %); 35 kN: 320 vs 268 MPa (19 %); 40 kN: 366 vs 282 MPa (30 %). At the preset curvature: 366.9 vs 282.3 MPa, 30.0 % high (the readout's Difference column shows 23 % because it divides by the flexure value; accept either if the base is stated). The error grows quickly beyond P_Y because M keeps rising while the outer-fiber stress barely does. (b) At 40 kN's curvature, EPP is 1.34 % above the material curve; the data follow the material curve (0.07 %). (c) With the strip made of aluminum the neutral axis moves from 24.00 to 28.58 mm (+4.57 mm), and at M = 1500 N·m the bottom stress would be predicted as 108.5 MPa instead of 189.7 MPa: ignoring

the strip underestimates the steel's stress by a factor of 1.75. (d) The RMS residual scales with κ (A6), so compatibility (Eq. 3) holds through Part C. What failed is Hooke's law: Eqs. 5–6 assume a linear σ–ε relation, and Eq. 12 assumes a flat plateau.

4. Challenge and Extension solutions

C1. Appendix D.6: $Z = bt_f(h - t_f) + t_w h_w^2/4$ = 238.6 × 10³ mm³, S = 209.8 × 10³ mm³, k **= 1.1371** (rectangle 1.5). The experiment (κ = 16.0 κ_Y) gives M/M_Y = 1.1368, essentially fully plastic. M_Y = 52.5 kN·m, M_p = 59.7 kN·m. In an I-section most of the material is already near $\pm\sigma_Y$ at first yield, so there is little left to mobilize; a rectangle's lightly stressed core can still pick up a lot. Design against collapse (plastic design) therefore gains 50 % for a rectangle but only about 14 % for an I-section; the I-section is still far more efficient per unit mass.

C2. Part B (elastic): errors N_s = 10: −8.17 × 10^{-3}, N_s = 20: −2.05 × 10^{-3}, N_s = 40: −5.17 × 10^{-4}; successive ratios 3.985 and 3.964 (4 for second order). Richardson from 20 and 40: relative error −6.21 × 10^{-6}. Part C with EPP: N_s = 10: −3.39 × 10^{-3}, N_s = 20: −3.36 × 10^{-4}, N_s = 40: −2.58 × 10^{-4}, ratios 10.1 and 1.3; the least-squares slope over N_s = 2–256 on the plot is 2.09. The midpoint rule is second order for a smooth integrand; the EPP stress has a kink at $\pm y_Y$, and the error contributed by the strip containing the kink depends on where in that strip the kink falls, so the error oscillates about an N_s^{-2} trend. Triangular load: N = 2: 6.25 × 10^{-2}, N = 4: 1.56 × 10^{-2}, N = 8: 3.91 × 10^{-3}, N = 16: 9.77 × 10^{-4}, N = 32: 2.44 × 10^{-4} (slope 2.00). Four-point bending: the shear is piecewise constant and the load points are nodes, so the trapezoid rule integrates it exactly at any N (largest error 0.0).

C3. (a) Every strain scales by $(1 + \delta)$, so A_0 and A_1 both scale and $y_{NA} = -A_0/A_1$ is unchanged while κ scales. (b) Replacing y_h by $y_h + d$ changes A_0 by $-A_1 d$ and leaves A_1 unchanged. (c) For one gage, $\delta A_1/A_1 = -\delta y_j(y_j - \bar{y})/\sum(y_i - \bar{y})^2$ to first order; with $\sum(y_i - \bar{y})^2$ = 2 288 mm², a 0.1 mm error changes κ by at most 0.089 %. At 20 kN it is equivalent to a 10.5 με strain error at that gage, which is how A2's residuals (1–3 με) bound the gage-height errors to a few hundredths of a millimeter. None of these explains a discrepancy here; the gage factor (a) is the largest term in A3.

X1. Strains $\varepsilon = \varepsilon_0 - \kappa y_h$ in both layers (compatibility, perfect bond); stresses $E(\varepsilon - \alpha\Delta T)$; $N = 0$ and $M = 0$ give two linear equations for ε_0 and κ. With measured dimensions and E: $\kappa/\Delta T$ **= −0.2114 × 10^{-3} m^{-1}/K**, hogging (concave down) because the aluminum on top expands more. At 50 K: κ = −0.0106 m^{-1} (radius 94.6 m); interface stresses steel +50.6 MPa and aluminum −23.8 MPa; steel bottom 36.8 MPa; aluminum top 12.9 MPa. *Check:* equal coefficients give $\kappa = 0$ and no stress. The interface shear needed to transfer these forces is concentrated near the ends, which is where bimetal bonds fail.

X2. Unloading from M_{max} leaves $\sigma_{res}(c) = -\sigma_Y + M_{max}c/I = \sigma_Y(M_{max}/M_Y - 1)$ on the compression side (tension), and the opposite on the tension side. On reverse loading by ΔM, the extreme fiber reaches $-\sigma_Y$ again (on the original tension side) when $\sigma_Y(1 - M_{max}/M_Y) - \Delta M c/I = -\sigma_Y$, i.e. after a total change of $2M_Y$ from M_{max}: reverse yield at $M = M_{max} - 2M_Y$. Since $M_{max} \le M_p = 1.5M_Y < 2M_Y$, unloading to zero never reverse-yields (the residual is at most 0.5 σ_Y). Reloading in the original direction is elastic up to M_{max} (shakedown; the same idea as the autofrettage of gun barrels and the presetting of springs).

X3. T-section from the I-beam preset (flange 100 × 10 mm, web 6 mm, depth 200 mm, area 2 140 mm²): elastic NA 141.73 mm above the bottom; plastic NA 178.33 mm (equal areas, in the web just below the flange); strip summation at $10^4\kappa_Y$ puts it at 178.33 mm with M/M_p = 1.00000. Z

= 112.5 × 10³ mm³, S (to the extreme fiber, at the bottom of the web) = 61.8 × 10³ mm³, k **= 1.819**. The elastic axis balances first moments of area; the plastic axis balances areas. They coincide only for a section symmetric about the neutral axis.

5. Design problem D1: reference solution

D1. $M_{max} = wL^2/8 + PL/4$ = **7.125 kN·m** at midspan; $\delta_{allow} = L/240$ = 12.5 mm. Steel: $\sigma_Y - U$ = 294.0 MPa, allowable 176.0 MPa, S_{req} = 40.47 × 10³ mm³, I_{req} = 2.332 × 10^6 mm^4 (E = 205 GPa). Aluminum: 271.7 MPa, 162.7 MPa, S_{req} = 43.79 × 10³ mm³, I_{req} = 6.991 × 10^6 mm^4 (E = 68.39 GPa, measured).

Table K3. Steel candidates (stress is the same for both materials).

Section	Mass (kg/m)	σ_{max} (MPa)	δ, steel (mm)	Acceptable
I-100	7.79	218.1	17.85	no
I-120	10.07	139.6	9.52	yes
I-140	12.60	94.9	5.55	yes
I-160	15.26	68.3	3.49	yes
I-180	18.30	50.4	2.29	yes
I-200	21.44	38.6	1.58	yes
RT-100x50x4	8.94	247.2	20.23	no
RT-120x60x5	13.38	138.2	9.42	yes
RT-150x100x5	18.89	70.8	3.86	yes
RT-200x100x6	27.20	39.7	1.63	yes

Table K4. Aluminum candidates.

Section	Mass (kg/m)	σ_{max} (MPa)	δ, aluminum (mm)	Acceptable
I-100	2.67	218.1	53.50	no
I-120	3.45	139.6	28.53	no
I-140	4.32	94.9	16.64	no
I-160	5.24	68.3	10.47	yes
I-180	6.28	50.4	6.87	yes
I-200	7.36	38.6	4.73	yes
RT-100x50x4	3.07	247.2	60.63	no
RT-120x60x5	4.59	138.2	28.24	no
RT-150x100x5	6.48	70.8	11.58	yes
RT-200x100x6	9.33	39.7	4.87	yes

Steel: I-120, 10.07 kg/m (30.2 kg, about $66); utilization 0.79 in stress and 0.76 in deflection, so the two requirements are nearly balanced. **Aluminum: I-160**, 5.24 kg/m (15.7 kg, about $102); utilization 0.42 in stress and 0.84 in deflection: deflection governs, because aluminum's E is a third of steel's while its strength is similar. The aluminum beam is about half the mass at about 1.5 times the cost; either is acceptable if justified. *Assumptions that could invalidate a design:* lateral–torsional buckling of the unbraced compression flange over 3 m (check bracing, or a buckling check that comes later in the course); local buckling of thin flanges (the catalog proportions are compact);

load placement other than at midspan; long-term creep is not an issue for these metals at room temperature.

Rubric detail (15 points total in the report). Moment and requirements (3). Strength basis (3.75): $\sigma_Y - U$ from the student's Lab 1 data and measured E for aluminum. Every requirement checked, governing one named for each material (3). Trade-off between at least two options with numbers (3.75). Presentation and assumption (1.5). Typical partial credit: w in kN/m mixed with mm (−2); deflection not checked (−4, it governs aluminum); handbook values instead of measured (−1); S and I swapped (−2).

6. Expected ranges for real class data

Table K5. Expected data ranges.

Quantity	Expected	If outside, suspect
RMS residual of the profile fit, Part A	1–5 με	a gage misplaced or debonded; wrong height recorded
y_{NA}, bar A	$h/2$ ± 0.3 mm	gage heights measured from the wrong face; bar not seated (twist)
E of 6061-T6 (GPa)	67–71	gage factor setting; h measured at the wrong place; a wrong
ν from the top-transverse gage	0.31–0.35	gage not centered or not transverse
V/P from the bottom gages	0.45–0.55	gage x positions; local effect of the load roller at x = 150 mm
Bar B NA (mm above bottom)	23.5–24.5	adhesive layer thickness; strip not 1018
Permanent curvature after Part C (m^{-1})	0.06–0.10	plateau of the particular bar; load overshoot

7. Common student mistakes

1. Measuring gage heights from the top instead of the bottom, or mixing y (from the neutral axis) with y_h (from the bottom).
2. Using $\sigma = Mc/I$ with c measured to the far fiber of the composite instead of using the transformed section and the factor n.
3. Forgetting that $\kappa = -A_1$: a positive slope of strain against height means hogging.
4. Fitting M against κ through the unloading steps or the zero step, or forcing the line through the origin without saying so.
5. Using P instead of $P/2$ in $M = (P/2)a$.
6. Computing E from a single bottom gage with the handbook neutral axis, which hides the plane-sections test.
7. Applying Eq. 6 in Part C and calling 366 MPa the stress in an aluminum bar whose tensile strength is about 314 MPa.
8. Taking the permanent curvature as $\kappa_{max} - \kappa_Y$.
9. In A5, comparing the steel stress with $-My/I_t$ without the factor n.
10. In D1, checking stress only, or using the steel modulus for the aluminum beam.

8. Grading rubric

Table K6. Grading rubric (100 points plus up to 10 Challenge points). Core analysis A1–A7 totals 45.

Part	Points	Full credit	Common deductions
Pre-lab	15	P1–P6 correct with units; P1 an estimate made before measuring	P2(c) missing (−1); P4 inversion wrong (−2)
Data & procedure	10	Tables 3–4 complete; checks in steps 1–6 recorded	gage heights not measured (−3)
A1	6	$M(x)$ from gages vs statics, V/P with U, intercept check	$P/2$ error (−2)
A2	7	profiles, κ, y_{NA} with U, residual discussion	no residuals (−3)
A3	7	E with budget, comparison with Lab 2, flexure check	no budget (−3)
A4	5	ν with U, ρ', sign check	—
A5	8	NA, EI, four stresses vs prediction, $\int \sigma\, dA$ check	factor n missing (−3)
A6	8	M–κ with both models, flexure overstatement, permanent curvature, residual stress	elastic formula past yield (−3)
A7	4	complete table, normalized figure, every discrepancy explained	unexplained discrepancy (−1 each)
M1	10	(a)–(d) with simulation numbers and the plane-sections argument	no simulation numbers (−2 each)
D1	15	see Sec. 5	see Sec. 5
Communication	5	labeled axes, units, $x \pm U$ format	each unlabeled axis (−1)
Challenge	+10	C1 (3), C2 (4), C3 (3)	—

Every Core answer must include the required check and interpretation; deduct 1 point per question where either is missing.

9. Instructor notes

Before the lab. Bars: 6061-T6, 25.4 × 50.8 mm (1 × 2 in) bar stock, 700 mm long. Bar A gets ten gages (five on one side face at 5, 15, 25.4, 35 and 45 mm, top, bottom and top-transverse at midspan, and two bottom gages at 100 and 150 mm from the left support); bar B gets eight. Bond the 6.35 mm 1018 strip to bar B with a structural epoxy (thin, uniform bond line; clamp while curing). Part C consumes one bar A per section run: either give each team its own bar (about 2.5 kg of 6061 bar per team) or run Part C once as a demonstration and share the data. Check the fixture's rollers turn freely; seized rollers add axial force and spoil the neutral axis.

Timing (2 h). 0:00–0:10 safety and pre-lab collection; 0:10–0:25 dimensions and gage heights; 0:25–0:55 Part A (eleven steps with holds); 0:55–1:20 Part B; 1:20–1:45 Part C; 1:45–2:00 straightedge check and simulation (step 8).

Safety. Demonstrate the overload stop and a controlled stop. Under 40 kN the 6061 bar deflects only a few millimeters, but a spreader that tips releases the load suddenly; insist on the centering check.

Running without equipment. Give students `sample_data.csv` and Appendix A of the handout; everything in this key applies unchanged. A good hybrid is to run Part A live (it is non-destructive) and use the sample data for Parts B and C.

Your own class data. Export the columns listed in Table A1 of the handout, one row per load step; fill a measurements JSON like `src/specimen_measurements.json`, with the class's Lab 1 and Lab 2 values in `reference`; run `src/lab04_analysis.py` and `src/make_key.py`.

The simulation. Students use it in P5, step 8, A6, M1, C1, C2 and X3. Its readout separates verification rows (closed form vs. integration or strip summation, "agree") from model-comparison rows ("different models"); the tests are in `tests/` and summarized in `tests/VERIFICATION.md`.

10. Self-critique and revisions

Before release, the lab was reviewed as a demanding instructor at a top program would review it. The three biggest weaknesses found, and the fixes made:

1. **The planned lab verified the flexure formula with a handbook modulus and an assumed neutral axis,** which is circular: it can only "confirm" what it assumes. *Fix:* seven gages through the depth measure the strain profile, so the neutral axis and curvature come from the data without assuming the material is elastic (Eq. 15). E is measured from moment and curvature and compared with Lab 2, a transverse gage measures ν (A4), and the residuals test plane sections directly (A2, M1(d)).
2. **Shear and moment diagrams (§5.1–5.2) were drawn but never measured.** *Fix:* two bottom gages in the shear span turn strain into $M(x)$ and the moment gradient into V (Eq. 16, A1), with an intercept check at the support. P2 adds the sensitivity to load-point placement.
3. **No test of where the formulas fail.** *Fix:* Part B tests the transformed section on a real two-material beam, including the equilibrium check $\int \sigma \, dA = 0$ from measured strains (A5). Part C loads past first yield and compares the measured moment–curvature curve with the EPP formula and with the material's own curve, quantifies the 29.8 % overstatement of Mc/I, and predicts the permanent curvature (agreement: 0.0827 vs 0.0826 ± 0.0028 m^{-1}).

Known limitations (not fixed). The truth model assumes a perfect bond and an elastic steel strip. It ignores the local contact stresses under the rollers, which affect the bottom gage at x = 150 mm (50 mm from a load roller) in a real test; extrusion residual stresses, which make the onset of yield in a real bar less sharp; and the gages' transverse sensitivity (about 1 %). Part C consumes a bar.